



2.2 Composite and inverse functions

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The composite function $(f \circ g)(x) = f(g(x))$ applies g first, then f , requiring the range of g to be compatible with the domain of f . The inverse function f^{-1} reverses the mapping of f ($f^{-1}(f(x)) = x$) and exists only if f is one-to-one (a bijection) on its domain; its graph is the reflection of $y = f(x)$ in the line $y = x$. To find f^{-1} , write $y = f(x)$, swap x and y , and solve for y .

Worked example: follow the reasoning

Given $f(x) = 2x + 3$, find $f^{-1}(x)$ and $(f \circ f)(1)$.

f^{-1} : let $y = 2x + 3$, swap: $x = 2y + 3$, solve: $y = (x - 3)/2$, so $f^{-1}(x) = (x - 3)/2$.

$(f \circ f)(1) = f(f(1)) = f(5) = 13$.

Quick check · multiple choice

If $f(x) = 3x - 1$ and $g(x) = x^2$, what is $(f \circ g)(2)$?

- A 5
- B 11
- C 35
- D 8

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Let $f(x) = (x + 2)/3$ for $x \in \mathbb{R}$. [7]
- (a) Find $f^{-1}(x)$.
 - (b) Verify that $(f \circ f^{-1})(x) = x$.
 - (c) Let $g(x) = x^2 - 1$ for $x \geq 0$. Find $(g \circ f)(4)$ and state why g must be restricted to $x \geq 0$ for g^{-1} to exist.
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Final answer: _____



Answers and explanation

Question 1 [7]

(a) $y = (x+2)/3 \rightarrow 3y = x+2 \rightarrow x = 3y-2$, so $f^{-1}(x) = 3x-2$.

(b) $f(f^{-1}(x)) = f(3x-2) = (3x-2+2)/3 = 3x/3 = x$. Verified.

(c) $f(4) = (4+2)/3 = 2$, $g(2) = 2^2 - 1 = 3$. g must be restricted because $g(x) = x^2 - 1$ is not one-to-one over all reals (e.g. $g(1) = g(-1)$), so it has no inverse unless the domain is restricted to make it one-to-one.

Quick check answer

B. $g(2) = 4$, then $f(4) = 3(4) - 1 = 11$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.